

# System Architecture & Data Flow

Smart Waste Intelligence Platform (SWIP) • Hackathon Software Demo

## 1. Multi-Tier Architecture

SWIP is built with a decoupled multi-tier architecture uniting client presentation, real-time event simulation, business logic APIs, and predictive decision engines.

## 2. Component Layers

- **Presentation Layer:** Unified Single Page Application hosting Admin Command Center, Citizen Portal, Smart Bins Monitor, AI Insights, and Worker Dispatch views.
- **Event & Simulation Layer:** Reactive event bus broadcasting simulated IoT sensor pulses (fill %, weight kg, temperature C) in real-time.
- **API & Logic Layer:** REST endpoints managing bin state, complaint lifecycles, worker shifts, and task dispatches.
- **AI & Optimization Layer:** Multi-factor prioritization algorithm and waste image classification pipeline.

## 3. Closed-Loop Flow

Simulate Bin Surge (95%) -> Event Ingestion -> AI Critical Detection (Score 94) -> Auto-Dispatch Task -> Worker Arrives & Empties -> Proof Upload -> Bin Telemetry Flush (18%) -> Dashboard KPI Update.